A BIOINFORMATICIAN'S QUARTO JOURNEY

LLM-Powered Clinical Reporting with Quarto

Validated analysis in, evidence-grounded report out — the LLM writes the report, not the statistics.

Speaker: Jaspreet Pabla

Date: 20/Oct/2026 · 10-minute session

LAST YEAR → THIS YEAR

Computation Is Reproducible. The Report Isn't.

LAST YEAR: SOLVED

70% faster report generation

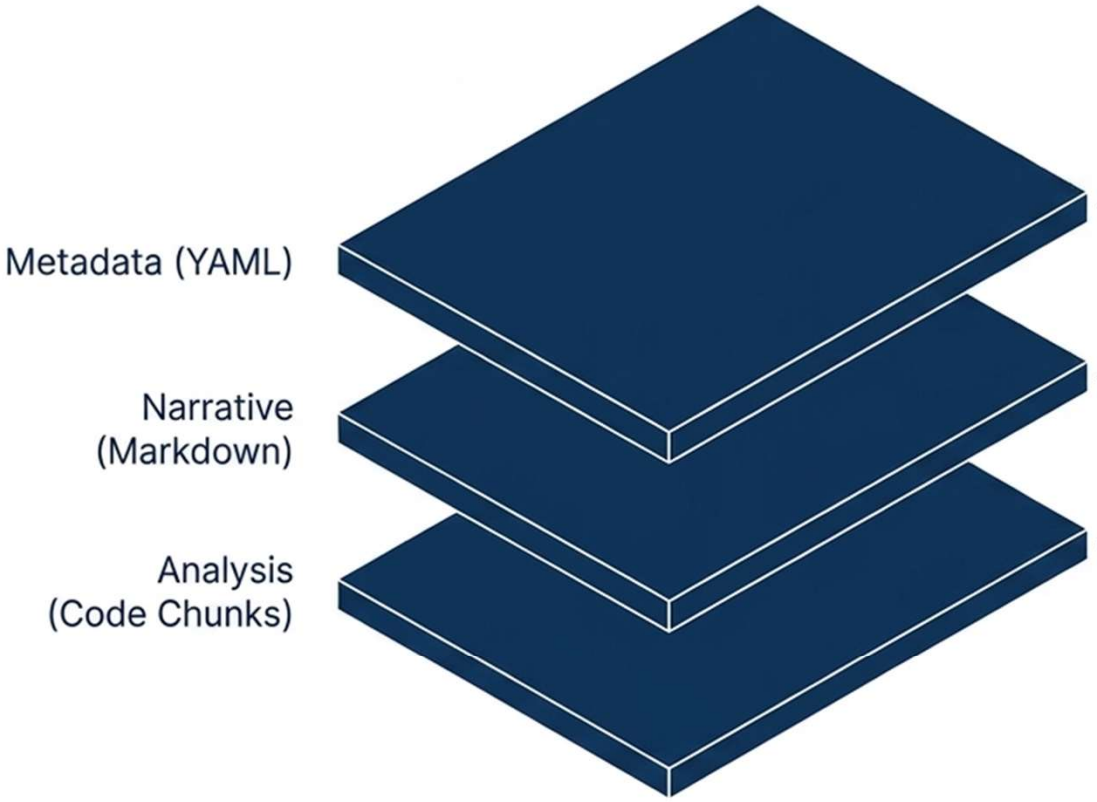
100% audit-ready traceability

1 doc R + Python + interactive

THIS YEAR: STILL MANUAL

- Methods and Results are written by hand, every study
- The sentence has no traceable link to the p-value it describes
- Two authors, two styles, for the same class of analysis

The Anatomy of Native Generation

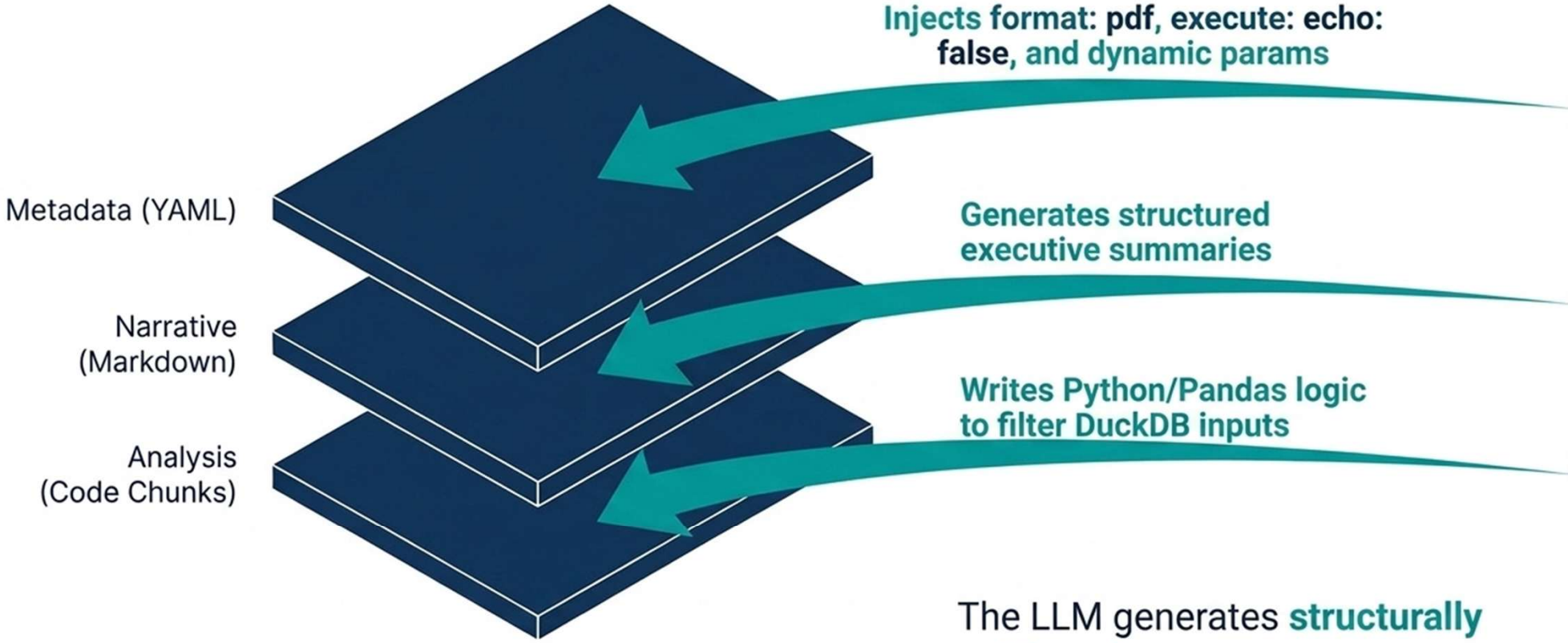


A true **co-author** doesn't just write words.

It configures the **metadata**, authors the **narrative**, and writes the **execution logic**.

Gemini Notebook

Injecting Intelligence at Every Layer

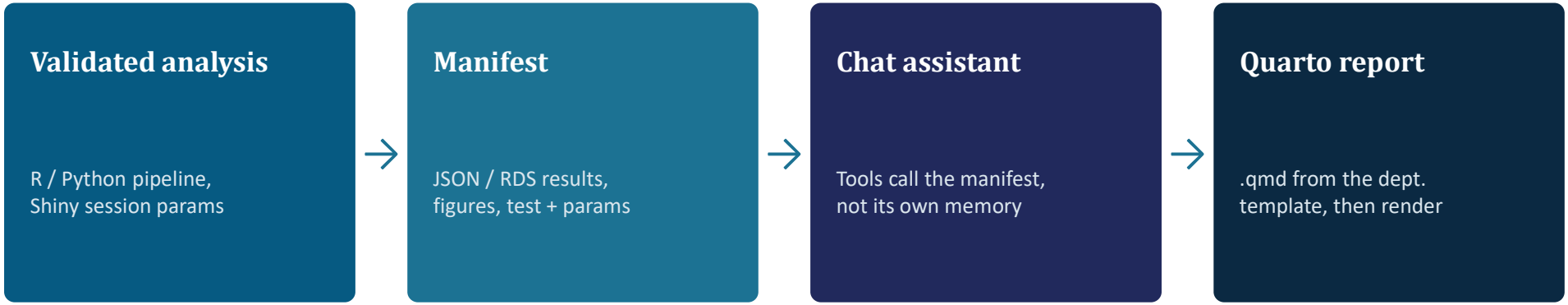


The LLM generates **structurally valid markup**, not just answers.

Gemini Notebook

OUR DESIGN

Chat Is the Interface, Not the Source of Truth



ASSISTANT TOOLS

`list_results`	`get_figure`	`add_section`	`set_logo`	`preview`	`render`

Every number in the report comes from a tool call on the manifest.

DEMO (ILLUSTRATIVE MOCK-UP)

Building a Report by Chatting

Add the PCA and ANOVA for batch 2, then describe the preprocessing.

Added 2 figures and 1 table from manifest run_0412. Preprocessing: log2 normalization, batch correction, 812 of 847 features retained.

Add our logo and preview.

Preview ready: 3 sections, 0 ungrounded values.

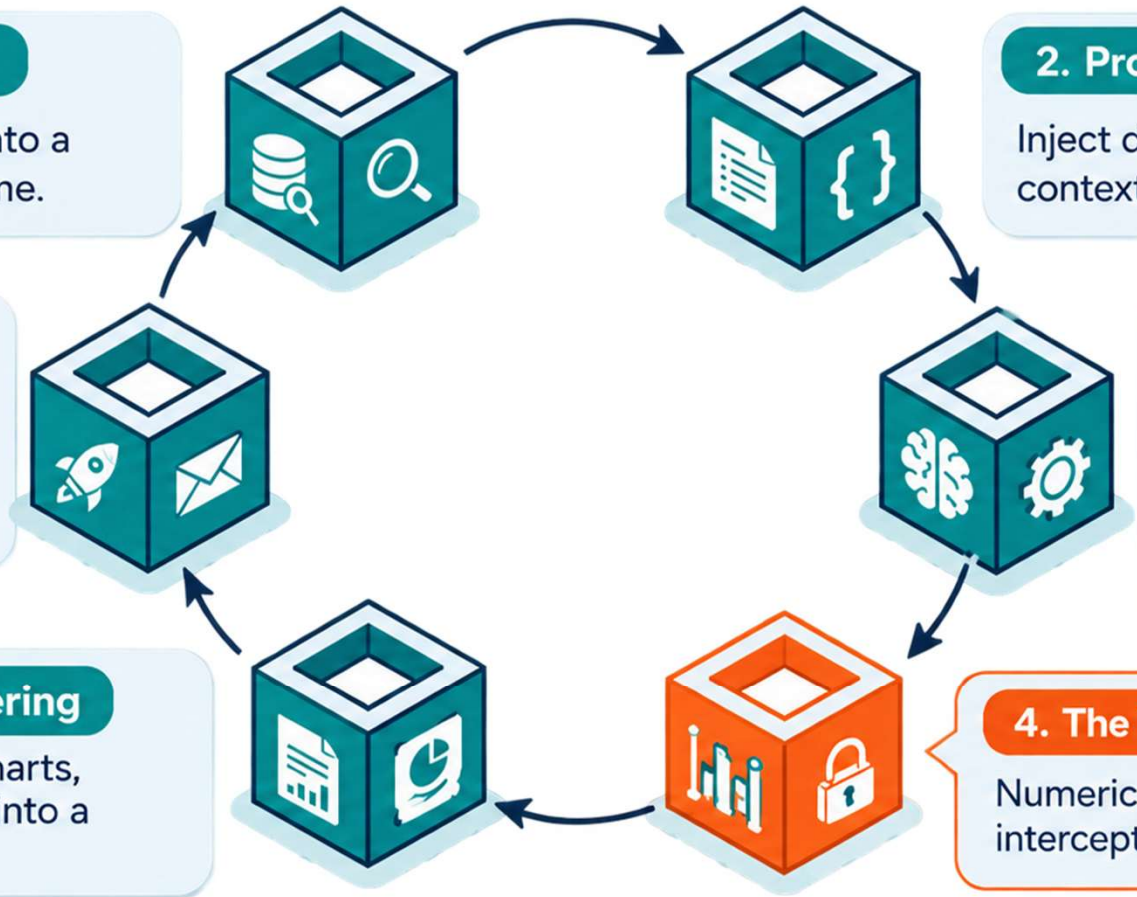
Metabolomics Analysis Report

Methods

Results

Author commentary — written by the scientist

The 6-Step Automated Pipeline



1. DuckDB Query

Aggregate raw data into a clean Pandas dataframe.

2. Prompt Construction

Inject data and historical context into Jinja templates.

3. Local Generation

Ollama generates the narrative (Temperature: 0.3 for analyst tone, Seed: 42 for reproducibility).

4. The Trust Gate

Numeric validation script intercepts output.

5. Quarto Rendering

Combine Python charts, data, and LLM text into a .qmd file.

6. Delivery

Dispatch via Slack webhook and email SMTP simultaneously.

EFFICIENCY WITH CONTROL

One Engine, Two Modes

Manual

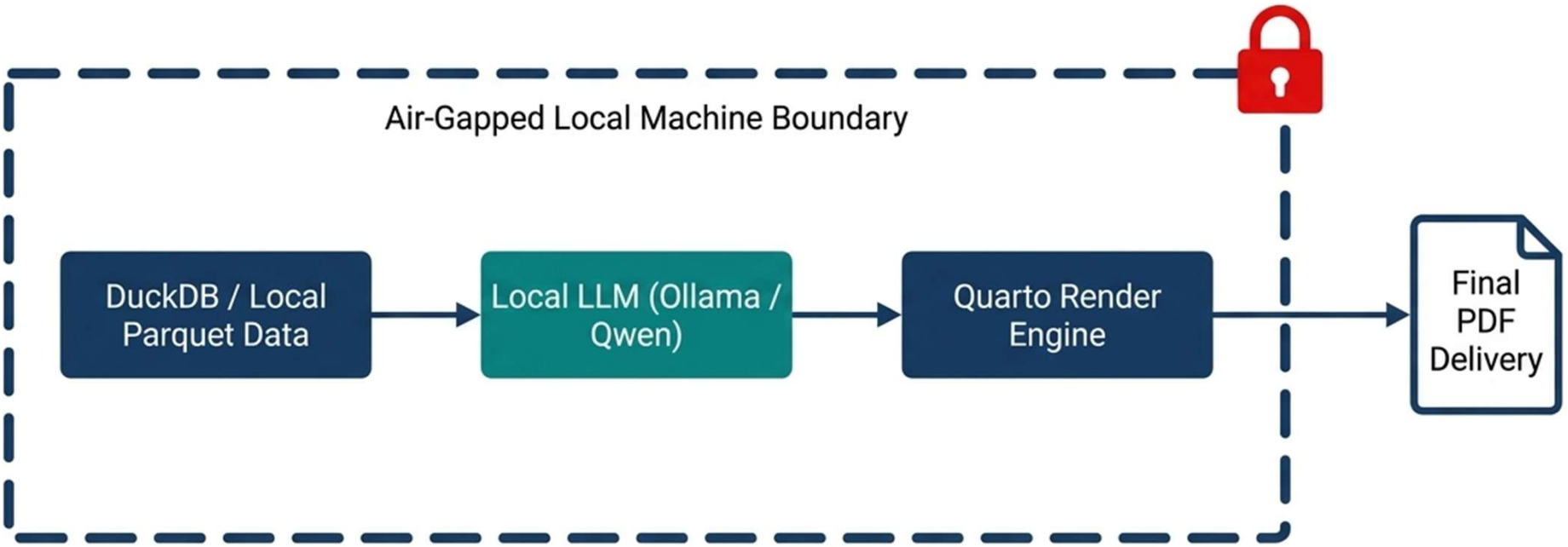
A scientist chats, adds sections, previews, edits. Best for new studies and one-off requests.

Automated

A stored recipe runs when a new manifest lands and drops a draft for review. Best for recurring reports.

Same tools · same manifest · same template · same human sign-off

Secure, On-the-Go Local AI



Zero data leakage: Revenue, churn, and account data never leave your hardware.

Ollama + DuckDB + Quarto: The ultimate private reporting stack.

Gemini Notebook

Thank You

Better reports. Faster. With trust.

LLM-powered clinical reporting with Quarto

Faster report drafting

AI-assisted composition reduces manual effort and speeds up delivery.

Data handling governed

Your data stays in your controls with secure deployment options.

Results grounded in validated analysis

Every number comes from trusted R/Python outputs, not the LLM's guesswork.

Human review before approval

Scientists add interpretation, ensure accuracy, and maintain oversight.

KEY TAKEAWAYS

- Reproducible analysis and reporting
- Secure and flexible deployment
- Verified statistical precision
- Human oversight at every step

Thank you!